

A Tiny, GSM-Based, Long-Range SPY DEVICE

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Bluetooth-based spying devices enable eavesdropping, much like those in James Bond films. However, Bluetooth and Wi-Fi have a limited range, restricting their effectiveness. Now, imagine discreetly intercepting conversations, uncovering criminal intent, and taking timely action—an intriguing possibility.

A compact GSM-based surveillance device can enable monitoring across vast distances, even between countries. Its small size ensures easy concealment while maintaining an unlimited range.

Strictly intended for security and surveillance, it must not be used for unauthorised purposes. The series of articles beginning with this one covers various long-range spying technologies, starting with GSM-based communication, followed by FM and Wi-Fi-based methods.

A compact board is required to construct this device. The Indus-Board Coin is an ideal choice,

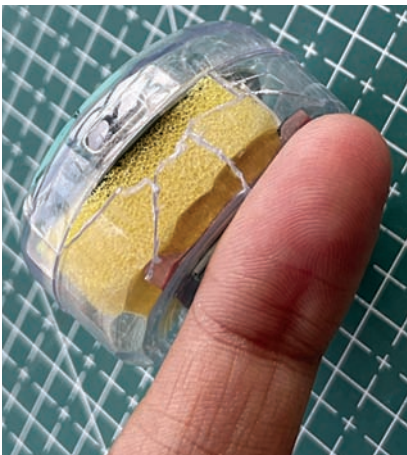


Fig. 1: Author's prototype

```

SPYdevice.ino
1  #include <HardwareSerial.h>
2  #include "Adafruit_FONA.h"
3
4  #define FONA_RX  43 // SIM800L RX connected to ESP32-S2 TX (Pin 43)
5  #define FONA_TX  44 // SIM800L TX connected to ESP32-S2 RX (Pin 44)
6  #define FONA_RST 5  // Reset pin (if needed)
7
8  // Initialize hardware serial
9  HardwareSerial fonaSerial(1); // Use UART1
10
11 // Initialize FONA
12 Adafruit_FONA fona = Adafruit_FONA(FONA_RST);
13
14 void setup() {
15   Serial.begin(115200);
16   Serial.println(F("FONA Spy Device Initializing..."));
17
18   // Start SIM800L Serial communication
19   fonaSerial.begin(4800, SERIAL_8N1, FONA_RX, FONA_TX);
20
21   if (!fona.begin(fonaSerial)) {
22     Serial.println(F("Couldn't find SIM800L"));
23     while (1);
24   }
25
26   Serial.println(F("SIM800L is ready"));
27
28   // Enable Caller ID Notification
29   if (fona.callerIdNotification(true, 0)) {

```

Fig. 2: The code snippet for pin definition

BILL OF MATERIALS

Components	Quantity	Description
IndusBoard Coin V1/V2	1	3cm development board
SIM800L module	1	2G GSM module for communication
Li-Po battery (3.7V) with BMS module	1	Li-Po battery with BMS module
Microphone module	1	Captures audio for transmission

Note: A UFL flexible GSM antenna can be attached to the device for better connectivity. In this case, it is not used as network connectivity is sufficient in the deployment area.

measuring only 3cm, with the SIM800L GSM module fitting perfectly within this small form factor. This combination allows the development of an efficient and discreet spying device capable of transmitting audio over long distances.

The required components are listed in the Bill of Materials table.

Software

Before writing the code, the required library must be installed. Follow these steps: